09/04/2026 | CN | Class

Shayan

"Compare CSMA/CA and CSMA/CD" assignment

minimum 5 page,
maximum 10 page
font size 12 pt

What is CSMA/CA?
CSMA/CD?

Then compare

— How to access Common Channel?

flow chart (?)

Advantages & Disadvantages

802.11 LAN Architecture

① Wireless host communicates with base station

base station = access point (AP)

BSS = Basic Service Set (aka "cell")

Contains —

1. Wireless Hosts
2. Access Point (AP) : base station
3. Ad hoc Mode : hosts only

802.11 : Channels, Association

① Spectrum divided into channels at different frequencies
Interference Possible : Channel can be same as that chosen by neighbouring AP.

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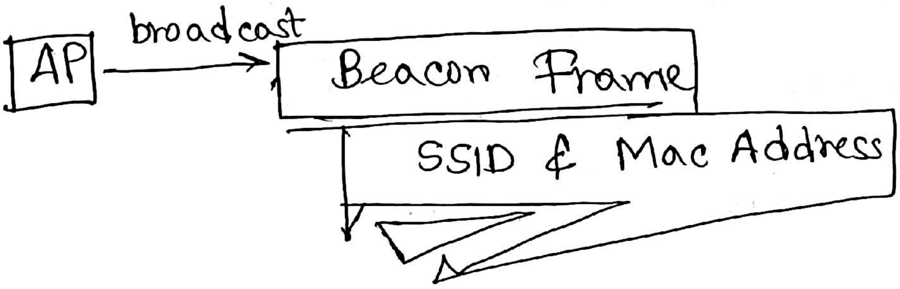
② Arriving host :
'Must associate with an AP'

- Scans channels, listening for beacon frames, containing AP's name (SSID) & MAC address.

- Selects AP to associate with
- then may perform authentication
- then typically run DHCP to get IP address in AP's subnet

"beacon frames"

→ active scanning
→ passive scanning



WEP / WPA / WPA2 / WPA3

Wireless Protected Access 1/2/3
Passphase (PSK)

passive ——→ important Shayon

① ~~active~~ scanning:

1. beacon frames sent from APs
2. association Request frame sent: H1 to selected AP
3. association Response frame sent: selected AP to H1

② ~~p~~ active scanning:

1. Probe req frame broadcase from H1
2. Probe res frames sent from APs
3. Association request frame sent: H1 to selected AP
4. Association response frame sent from selected AP to H1

CA → Collision ~~Average~~ Avoidance.

CD → Collision ~~Division~~ Detection

Carrier Sense Multiple Access

Avoid Collisions : 2+ Nodes transmitting at same time

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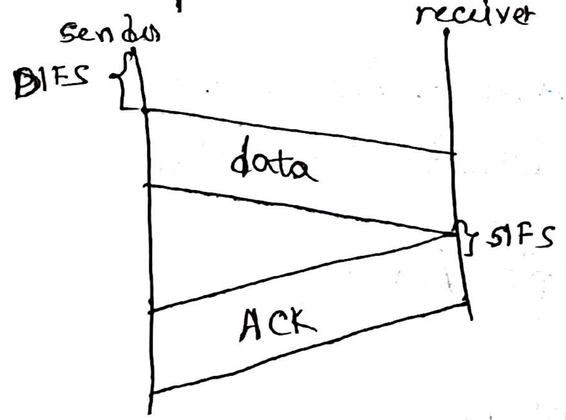
IEEE 802.11 MAC Protocol : CSMA/CA

802.11 Sender

1. If sense channel idle for DIFS then transmit entire frame (no CD)

2. If sense channel busy then start random backoff time timer counts down while channel idle transmit when timer expires if no ACK, increase random backoff interval. repeat 2.



802.11 receiver

if frame received OK

return ACK after SIFS (ACK needed due to hidden terminal problem)

CSMA/CA wired
CSMA/CD wireless

Avoidance Collisions

RTs = Request to Send

RTS & CTS

RTS - CTS exchange

Frame format → self study

Threshold value বইতে দেইখা নিও ☺

Frame too small? RTS & CTS not needed

802.11 frame: Addressing

2	2	6	6	6	2	6	0-2312	4
frame control	duration	address 1	address 2	address 3	Seq control	address 4	payload	CRC

Address 1: MAC address of wireless host/AP to receive this frame

Address 2: M " " " " " transmitting " "

Address 3: " " " router interface to which AP is attached

Address 4: used only in Ad-hoc Mode.

WiMAX $\boxed{\times}$

Link Algo }
DV Algo } next class